

3.2-3.3 Exam-Style Questions

Edexcel A-Level Economics · Theme 3 · Business objectives, revenues, costs and profits | Name:

_____ Date: _____

These questions are written for practice in the style of the exam. They are not past paper questions. Attempt them under timed conditions (about a minute per mark). Indicative mark scheme points are on the last pages.

1. Explain why marginal revenue is below average revenue for a firm facing a downward-sloping demand curve. (4)

2. Using a diagram, explain how a firm finds its profit-maximising output and the profit it makes. (6)

3. A supermarket chain plans to double the number of stores it operates. Examine two internal economies of scale it might gain. **(8)**

[illegible]

4. A small manufacturer earns average revenue of £6 per unit. Its average variable cost is £5.50 and its average cost is £7. Assess whether the firm should stop producing immediately. (10)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

5. Discuss whether the managers of a large public company are more likely to pursue sales maximisation than profit maximisation. (12)

Write on your own paper. Plan first: two lines of argument, one evaluation point for each, and a judgement.

6. Evaluate the view that a firm that grows larger will always benefit from lower average costs. (25)

Write on your own paper in 30 minutes. Structure: definitions, analysis, evaluation, analysis, evaluation, judgement.

Mark scheme points

Indicative content. Credit any valid point that is developed and applied.

Question 1 (4)

- A firm facing a downward-sloping demand curve must cut price to sell more (1)
- The lower price applies to all units sold, not just the extra one (1)
- So the extra revenue from the additional unit is less than the price (1)
- Therefore MR is below AR, and falls faster (twice as steep for a straight-line demand curve) (1)

Example: selling 5 units at £10 earns £50. To sell a sixth the price falls to £9, so revenue is $6 \times 9 = £54$. MR is £4, below the £9 AR.

Question 2 (6)

- Diagram with correctly labelled axes, AR, MR, MC and AC, with MC cutting AC at its minimum (2)
- Profit-maximising output Q^* is where $MC = MR$, marked with dashed lines (1)
- Explanation: below Q^* , $MR > MC$ so extra output adds to profit; above Q^* , $MC > MR$ so extra output reduces profit (1)
- Price P^* read from the AR curve at Q^* , and cost per unit C^* from the AC curve (1)
- Profit = $(P^* - C^*) \times Q^*$, shaded and labelled (1)

Question 3 (8)

Award analysis and application. Each developed economy up to 4 marks (knowledge 1, application 1, analysis 2).

- **Purchasing:** larger orders from suppliers earn bulk discounts, so the cost per item falls, so unit costs fall as the chain grows. Apply: a supermarket buys huge volumes of food.
- **Technical:** larger distribution centres and logistics systems spread the fixed cost of technology, so average costs fall.
- **Marketing:** national advertising costs are spread over more stores and sales.
- **Managerial:** specialist buyers and finance staff become affordable.
- **Financial:** lower interest rates on borrowing for larger firms.

Award at most 4 for each economy, and cap at the lower level if the chain of reasoning is not developed or has no application.

Question 4 (10)

For continuing (analysis): AR (£6) is above AVC (£5.50), so each unit covers its variable cost and contributes £0.50 towards fixed costs. Fixed costs must be paid whether it produces or not, so shutting down would lose the whole fixed cost, a bigger loss than continuing.

For stopping (analysis): AR (£6) is below AC (£7), so it makes a loss of £1 per unit. If this continues, it cannot cover all costs.

Evaluation: the short-run rule is $AR \geq AVC$, so the firm should not stop immediately; but in the long run it must leave unless it raises AR or lowers AC. Other points: whether the price rise is likely, when fixed contracts (rent, loans) end, cost of restarting, whether the low price is temporary, alternative uses of the factory, and the effect on workers and suppliers.

Judgement: the firm should continue in the short run, as long as it expects to improve AR or cut costs before its fixed commitments end, otherwise it should exit in the long run.

A one-sided answer caps evaluation. Look for a diagram with AR, AVC and AC as a bonus, not required.

Question 5 (12)

Analysis for sales maximisation: managers' pay, status and job security often rise with the size of the firm rather than with profit, so they may push for higher sales ($AC = AR$), which gives higher output and a lower price at normal profit. The separation of ownership and control (principal-agent problem) makes this possible.

Analysis for profit maximisation: shareholders want profit, and managers may be paid in shares or bonuses tied to profit, so they aim for $MC = MR$.

Evaluation: shareholder pressure and the threat of takeover limit how far managers can depart from profit; the level of monitoring; how easy it is to measure MC and MR; some managers satisfice; the objective can change over time with market conditions; the firm may pursue sales in the short run for market share, to earn profit later.

Judgement: a supported view on which is more likely, weighing ownership structure, incentives and monitoring.

Question 6 (25)

Definitions: average cost, economies and diseconomies of scale, the long run.

Analysis 1: growth lowers costs. Chains through internal economies (purchasing, technical, managerial, financial, marketing, risk-bearing) to lower LRAC, with a diagram of the falling part of the LRAC. Possibly external economies for the industry.

Evaluation 1: depends on whether output is high enough to reach MES; whether savings are real or spread across long chains; short run versus long run, since costs of expansion come first; the type of industry.

Analysis 2: growth can raise costs. Diseconomies of scale: communication, coordination, motivation and control problems, so LRAC rises beyond the MES. External diseconomies such as congestion and higher wages.

Evaluation 2: good management, technology and delegation can delay diseconomies; the size at which they appear varies by industry; firms may split into divisions.

Judgement: "always" is too strong. Growing larger lowers average cost up to the MES and beyond that depends on managing diseconomies. The answer should weigh which is more likely for the type of firm and time horizon, and use extract evidence if provided.